

Project Design Phase: Proposed Solution Template

Header Parameters

Parameter	Information
Date	02 October 2026
Team ID	
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Proposed Details

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Traditional role-based access control (RBAC) in enterprise platforms like ServiceNow provides static table or field-level permissions, which often fail when data access policies require dynamic evaluation based on runtime record values (e.g., restricting access based on a specific department, branch, or field status). This creates vulnerability to unauthorized data viewing, creation, modification, or

		deletion across user interfaces and API endpoints.
2.	Idea / Solution description	Implement Advanced Script-Controlled Access Control Lists (ACLs) on custom/target tables (such as u_institution_details) in ServiceNow. The solution dynamically evaluates record-level field values (e.g., current.branch == 'EEE') alongside specialized user roles (bb1 for Read, bb2 for Create, bb3 for Write, and bb4 for Delete) directly within server-side scripts (gs.hasRole()), ensuring end-to-end security across forms, lists, and REST/SOAP APIs.
3.	Novelty / Uniqueness	Unlike standard UI-level restrictions (Client Scripts/UI Policies) which can be bypassed via direct URL/API requests, or database splitting which creates unnecessary architectural overhead, this approach integrates dynamic conditional logic directly into ServiceNow's security evaluation engine. It delivers context-aware CRUD security with minimal execution latency (<1.8ms) without modifying underlying database

		schemas.
4.	Social Impact / Customer Satisfaction	Significantly enhances data privacy and governance compliance by enforcing strict least-privilege security policies. Organizations can guarantee that multi-departmental or sensitive institutional data is strictly compartmentalized, preventing internal data leaks and increasing stakeholder/customer trust.
5.	Business Model (Revenue Model)	Offered as a reusable security framework/module or custom application configuration package for enterprise IT Service Management (ITSM) and custom ServiceNow applications. Revenue can be generated through consulting, implementation services, custom script auditing, and ongoing security maintenance subscriptions.
6.	Scalability of the Solution	Highly scalable across any ServiceNow table or custom enterprise application module. Because security logic is executed natively on the ServiceNow platform server engine, it easily

		scales to handle thousands of concurrent users, large datasets, and diverse departmental role hierarchies without performance degradation or additional database overhead.
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